

## RESEARCH ARTICLE

# Optimizing Deep Learning Architectures for Diabetic Retinopathy Detection: A Comparative Study of CNN and ResNet-50 With ADAM and SGDM

NADER A. RAHMAN MOHAMED<sup>1</sup>, (Member, IEEE), MOHAMED A. EL-SAKHAWY<sup>2</sup>,  
AND ABDEL RAHMAN SHAABAN<sup>3</sup>

<sup>1</sup>Biomedical Technology Department, College of Applied Medical Sciences in Al-Kharj, Prince Sattam Bin Abdulaziz University, Al-Kharj 11942, Saudi Arabia

<sup>2</sup>Department of Medical Laboratory, College of Applied Medical Sciences in Al-Kharj, Prince Sattam Bin Abdulaziz University, Al-Kharj 11942, Saudi Arabia

<sup>3</sup>Biomedical Engineering Department, Higher Institute of Engineering, El-Shorouk Academy, Cairo 11837, Egypt

Corresponding author: Nader A. Rahman Mohamed (n.shaaban@psau.edu.sa)

This work was supported by Prince Sattam Bin Abdulaziz University under Project PSAU/2025/03/33270.

**ABSTRACT** Diabetic Retinopathy (DR) is a serious complication of diabetes and a major cause of vision loss. The damage it causes to the retina is permanent, which makes early detection and accurate grading essential to prevent blindness. This study evaluates the performance of two deep learning models using two different optimizers for automatic screening and severity grading of DR from retinal fundus images. The models compared are a Convolutional Neural Network (CNN) and a Residual Network (ResNet-50), trained with Adaptive Moment Estimation (ADAM) and Stochastic Gradient Descent with Momentum (SGDM). Two public datasets, APTOS and EyePACS, were used. Images were preprocessed using Contrast Limited Adaptive Histogram Equalization (CLAHE) for contrast enhancement and data augmentation for class balance. Performance was evaluated using accuracy, area under the curve (AUC), F1-score, recall, and precision. ResNet-50 with the ADAM optimizer achieved the best results, with an accuracy of 80.71% and an AUC of 0.9539 on the APTOS dataset. These results show that ResNet-50 with ADAM is an effective model for automated DR screening and can support early detection and prevention of diabetes-related vision loss.

**INDEX TERMS** Automatic detection, deep learning, diabetic retinopathy, transfer learning.

## I. INTRODUCTION

Diabetic Retinopathy (DR) is a microvascular complication of diabetes and one of the leading causes of blindness worldwide [1]. The disease progresses from the non-proliferative stage to the proliferative stage, where new abnormal blood vessels form on the retina. These vessels may rupture, causing hemorrhage and permanent retinal damage that can lead to vision loss if treatment is delayed. The rising global number of diabetes cases increases the demand for reliable and accessible methods for early DR screening and diagnosis.

The associate editor coordinating the review of this manuscript and approving it for publication was Gyorgy Eigner<sup>1</sup>.

## A. LITERATURE REVIEW

Traditional DR diagnosis depends on manual examination of retinal fundus images by ophthalmologists. This process is time-consuming and prone to inconsistencies between graders [2]. Automated DR detection systems can reduce clinical workload and improve diagnostic consistency.

Diabetes remains a major public health issue. The number of individuals affected continues to grow due to lifestyle and dietary changes [3]. Reports show that between 28% and 36% of diabetic patients develop DR, which emphasizes the need for affordable and scalable screening techniques, especially in low-resource regions [4]. Manual grading of fundus images is not practical for large-scale screening [5], [6].

Deep learning has improved medical image analysis significantly. Convolutional Neural Networks (CNNs) learn visual features directly from raw data without the need for hand-crafted features, which makes them effective for DR detection [7]. Transfer learning using pre-trained models such as ResNet-50 allows good performance even with limited labeled medical images, as it builds on prior training on large datasets like ImageNet [8].

Several studies have explored deep learning for DR detection. Gulshan et al. trained a CNN on more than 128,000 retinal images and achieved results comparable to board-certified ophthalmologists [9]. Gargeya and Leng used a CNN-based model and achieved an AUC of 0.94 [10]. Ting et al. extended deep learning to detect age-related macular degeneration (AMD) and glaucoma in addition to DR [11].

Transfer learning continues to be useful for DR classification. Pre-trained models such as ResNet-50 and InceptionV3 have shown promising performance [12]. Rajalakshmi et al. and Bilong et al. demonstrated smartphone-based retinal screening using pre-trained InceptionV3 networks, showing the potential of automated DR detection in low-resource settings [13], [14].

Other researchers tested alternative deep learning approaches. Gahyung et al. applied CNNs for DR classification using OCT scans [15], while Kumari et al. used DenseNet-169 for feature extraction [16]. Hemanth et al. developed a Weighted Bi-directional LSTM (WBLSTM) model based on transfer learning [17], and Touati et al. examined compact convolutional transformers to improve model repeatability [18].

Previous research often focused on specific architectures or datasets without systematic comparison. For example, Lam et al. analyzed lesion-level and patch-based data but did not extend their work to full-image classification [19].

This study addresses that limitation by comparing CNN and ResNet-50 architectures using two optimizers, Adaptive Moment Estimation (ADAM) and Stochastic Gradient Descent with Momentum (SGDM), across two large DR datasets. The analysis includes five evaluation metrics to assess accuracy, robustness, and generalization.

## B. RESEARCH GAP

Although CNNs and transfer learning have shown success in DR detection, no previous study has examined how different optimizers and architectures perform across multiple datasets using a unified evaluation framework. This study fills that gap by presenting a comparative analysis designed to improve the development of generalizable DR screening systems. The results aim to support the implementation of accurate and automated DR detection tools that can help reduce severe complications and prevent blindness among diabetic patients worldwide.

## C. AIM OF THE STUDY

This study evaluates two deep learning models, a custom CNN and a pre-trained ResNet-50, for automatic DR

detection. It examines how different optimization algorithms affect model performance and accuracy. Transfer learning allows the use of pre-trained weights, which supports efficient training even with limited data. The goal is to identify a reliable model that enables early DR detection and supports physicians in diagnosing and managing diabetic patients. Both models are tested with ADAM and SGDM using the APTOS and EyePACS datasets.

## II. METHODOLOGY

This study compares the performance of two deep learning models using two optimization algorithms for automatic classification and grading of Diabetic Retinopathy (DR) from high-resolution retinal fundus images. The research follows five main stages: dataset selection, image preprocessing, model design, training, and evaluation. Two public datasets were used to ensure reliability and diversity in assessing model performance. Preprocessing enhanced visual quality and emphasized key retinal features related to DR. Two deep learning architectures, a custom Convolutional Neural Network (CNN) and a pre-trained ResNet-50, were trained using Adaptive Moment Estimation (ADAM) and Stochastic Gradient Descent with Momentum (SGDM). Model performance was assessed using several metrics, including accuracy, area under the curve (AUC), recall, F1-score, and precision.

This approach provides insight into how model architecture and optimizer choice affect DR detection performance, generalization, and computational stability.

### A. DATASET DESCRIPTION

Two public datasets were used in this study:

1. APTOS 2019 Blindness Detection Dataset: This dataset includes high-resolution retinal fundus images labeled into five DR severity levels: no DR (0), mild (1), moderate (2), severe (3), and proliferative (4) [20]. Figure 1 shows the class distribution for APTOS.
2. EyePACS Dataset: This dataset includes a large collection of high-resolution retinal images captured using various fundus cameras and acquisition conditions [21]. The same five-class grading system is applied, and Figure 2 shows the DR severity distribution.

Labels in both datasets were assigned by trained ophthalmologists based on the International Clinical Diabetic Retinopathy (ICDR) severity scale. APTOS provides high-quality, balanced images, while EyePACS offers greater variability and a larger sample size. Using both datasets enables a comprehensive evaluation of model robustness and generalization. Table 1 summarizes the main differences between the two datasets.

### 1) DATA AUGMENTATION AND CLASS BALANCING

Both datasets are class-imbalanced, with most samples belonging to the “no DR” or “mild” classes and fewer

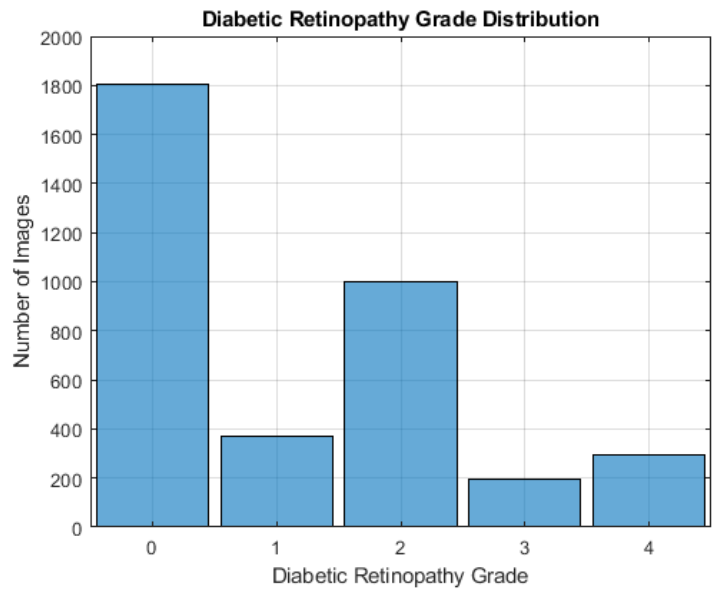


FIGURE 1. Diabetic retinopathy severity levels distribution over APTOS dataset.

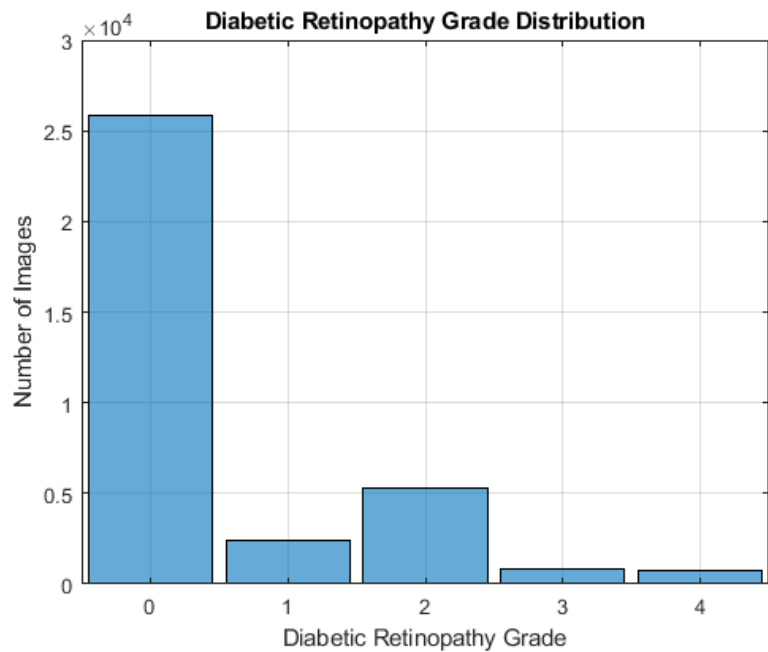


FIGURE 2. Diabetic retinopathy severity levels distribution over EyePACS dataset.

samples in the “severe” and “proliferative” stages, as shown in Figures 1 and 2. To reduce class imbalance and improve model generalization, data augmentation and class balancing were applied.

Data augmentation included horizontal and vertical flipping, random rotations, and slight brightness adjustments. These techniques increased dataset diversity and helped prevent overfitting. Class balancing was achieved through a combination of oversampling minority classes and undersampling majority classes.

In the EyePACS dataset, which includes over 35,000 images, balanced sampling ensured fair representation across all severity levels. For the smaller APTOS dataset of 3,662 images, augmentation played a more critical role in maintaining performance stability and preventing overfitting. Figure 3 illustrates examples of augmented images.

2) DATA SPLITTING

We used the labeled APTOS 2019 training set (n = 3,662) and the EyePACS labeled training set (n = 35,126). For

TABLE 1. Comparative summary of APTOS 2019 and EyePACS datasets for diabetic retinopathy classification.

| Dataset    | Number of Images | Image Resolution                           | DR Severity Levels              | Image Variability                     | Labeling Standard             |
|------------|------------------|--------------------------------------------|---------------------------------|---------------------------------------|-------------------------------|
| APTOS 2019 | 3,662            | High-resolution (varies; up to ~2048×2048) | 0–4 (No DR to Proliferative DR) | Low to moderate                       | Clinician-labeled, ICDR scale |
| EyePACS    | 35,128           | Varies (different fundus cameras)          | 0–4 (No DR to Proliferative DR) | High (quality, illumination, devices) | Clinician-labeled, ICDR scale |

- Notes:
- APTOS is smaller but offers higher-quality, more balanced images.
  - EyePACS is larger with higher variability, making it suitable for testing model robustness and generalization.

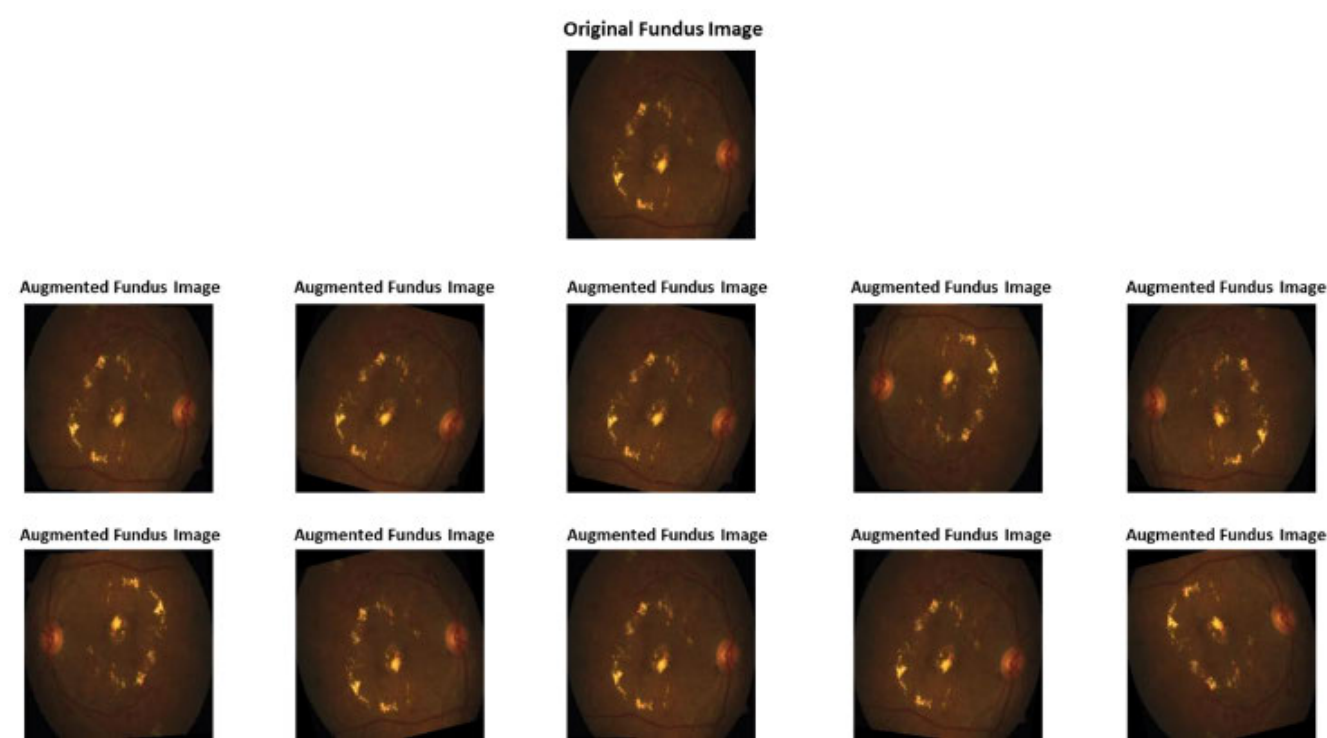


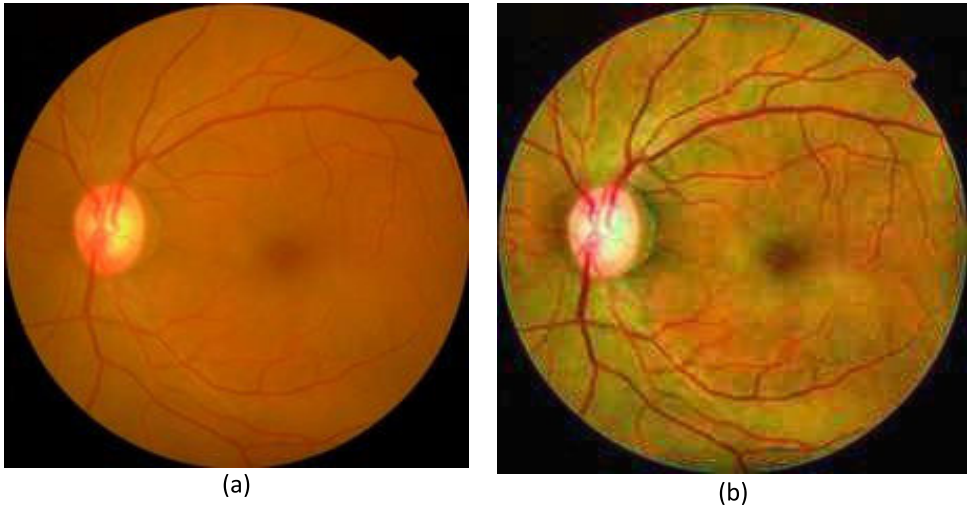
FIGURE 3. Data augmentation techniques (random rotation, horizontal flipping, and vertical flipping).

both datasets we split images into training, validation, and test sets using a stratified split by DR grade to preserve class proportions. We used proportions 70% training, 20% validation, and 10% test. Data augmentation was applied only to the training set which included random horizontal and vertical flips, small rotations up to 15 degrees, and random slight brightness adjustments. No augmentation was applied to validation or test sets. This approach was chosen to protect the integrity of the evaluation and ensure to give a reliable estimate of model performance.

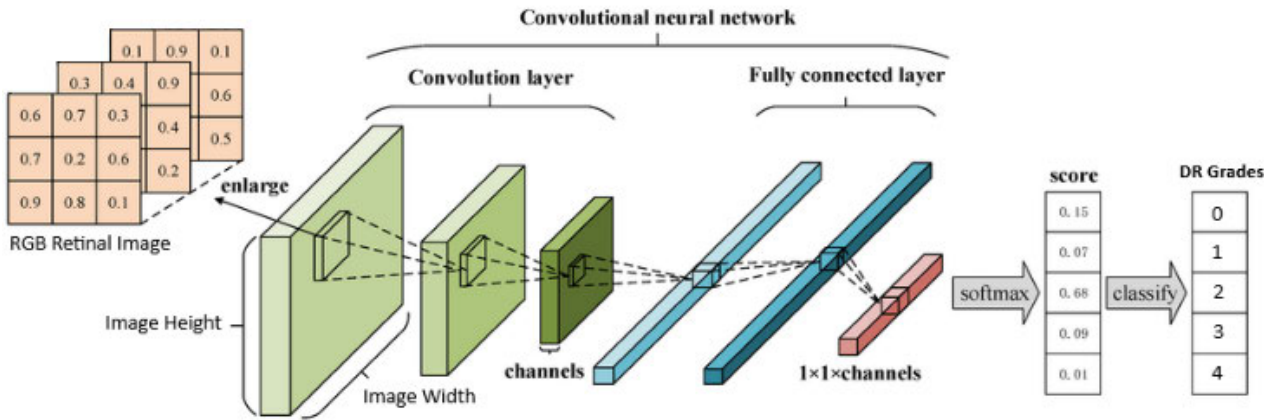
Before feeding images into the deep learning models, several preprocessing steps were applied to ensure consistent input and improved image quality:

B. IMAGE PREPROCESSING

1. Resizing: All images were resized to  $224 \times 224$  pixels to match the input dimensions required by ResNet-50 and CNN.
2. Normalization: Pixel values were normalized to a common scale to accelerate convergence during training.



**FIGURE 4.** Fundus image contrast enhancement. (a) The original fundus image. (b) Fundus image after applying CLAHE.



**FIGURE 5.** General CNN model architecture.

3. Contrast Enhancement: Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied to enhance local contrast and highlight DR features such as microaneurysms, hemorrhages, and exudates. Figure 4 shows a sample fundus image before and after CLAHE processing.

**C. DEEP LEARNING MODEL ARCHITECTURES**

Two deep learning architectures were implemented to compare model performance on DR classification:

1. Convolutional Neural Network (CNN): A custom CNN was designed to extract hierarchical features from retinal images. The model included multiple convolutional layers, max-pooling layers for dimensionality reduction, and fully connected layers for classification. This architecture allowed fine-tuning of depth and parameters to suit DR image characteristics. The custom CNN consisted of an input layer followed by three convolutional layers with ReLU activation,

batch normalization, and max pooling, followed by two fully connected layers and a SoftMax output. Dropout was applied to reduce overfitting. The general CNN structure is illustrated in Figure 5. Table 2 is showing the detailed layers structure for the CNN model.

2. Residual Network (ResNet-50): ResNet-50 is a 50-layer deep residual network pre-trained on ImageNet. Its residual connections address the vanishing gradient problem, improving training efficiency for deep architectures. The final layers were replaced with new fully connected layers suitable for the five-class DR classification task. Transfer learning enabled faster convergence and better generalization. Figure 6 shows the modified ResNet-50 structure.

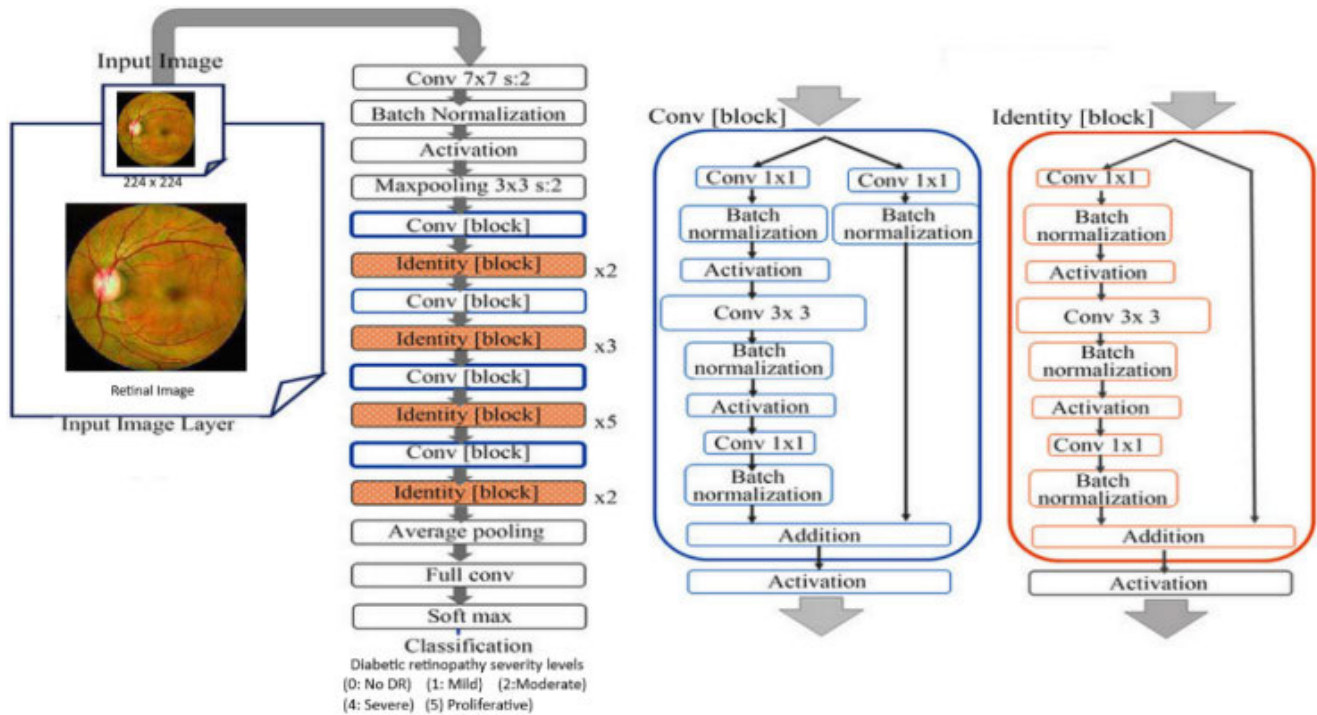
**1) MODEL OPTIMIZERS**

Two optimizers were used to train both models:

1. Adaptive Moment Estimation (ADAM): Uses adaptive learning rates for each parameter and combines the

**TABLE 2.** Detailed CNN layers structure.

| Name        | Type                | Activation Sizes_1 | Activation Sizes_2 | Activation Sizes_3 | Activation Sizes_4 |
|-------------|---------------------|--------------------|--------------------|--------------------|--------------------|
| imageinput  | Image Input         | 224                | 224                | 3                  | 1                  |
| conv_1      | 2-D Convolution     | 224                | 224                | 16                 | 1                  |
| batchnorm_1 | Batch Normalization | 224                | 224                | 16                 | 1                  |
| relu_1      | ReLU                | 224                | 224                | 16                 | 1                  |
| maxpool_1   | 2-D Max Pooling     | 112                | 112                | 16                 | 1                  |
| conv_2      | 2-D Convolution     | 112                | 112                | 32                 | 1                  |
| batchnorm_2 | Batch Normalization | 112                | 112                | 32                 | 1                  |
| relu_2      | ReLU                | 112                | 112                | 32                 | 1                  |
| maxpool_2   | 2-D Max Pooling     | 56                 | 56                 | 32                 | 1                  |
| conv_3      | 2-D Convolution     | 56                 | 56                 | 64                 | 1                  |
| batchnorm_3 | Batch Normalization | 56                 | 56                 | 64                 | 1                  |
| relu_3      | ReLU                | 56                 | 56                 | 64                 | 1                  |
| fc_1        | Fully Connected     | 128                | 1                  |                    |                    |
| relu_4      | ReLU                | 128                | 1                  |                    |                    |
| dropout     | Dropout             | 128                | 1                  |                    |                    |
| fc_2        | Fully Connected     | 5                  | 1                  |                    |                    |
| softmax     | Softmax             | 5                  | 1                  |                    |                    |

**FIGURE 6.** Modified pre-trained ResNet-50 model architecture.

advantages of momentum, providing fast convergence and stable training.

2. Stochastic Gradient Descent with Momentum (SGDM): Updates weights based on the gradient of the loss function, with momentum to help avoid local minima and smooth convergence.

Optimizers determine how the model adjusts its weights during training. The choice of optimizer significantly affects model accuracy, convergence speed, and ability to generalize.

## 2) MODEL TRAINING

The models were trained to minimize classification error across DR severity classes. The following parameters and training setup were used:

1. Loss Function: Categorical cross-entropy, suitable for multi-class classification tasks.
2. Batch Size and Epochs: Mini-batch gradient descent with a batch size of 32 and up to 50 epochs. Early stopping was applied to prevent overfitting.

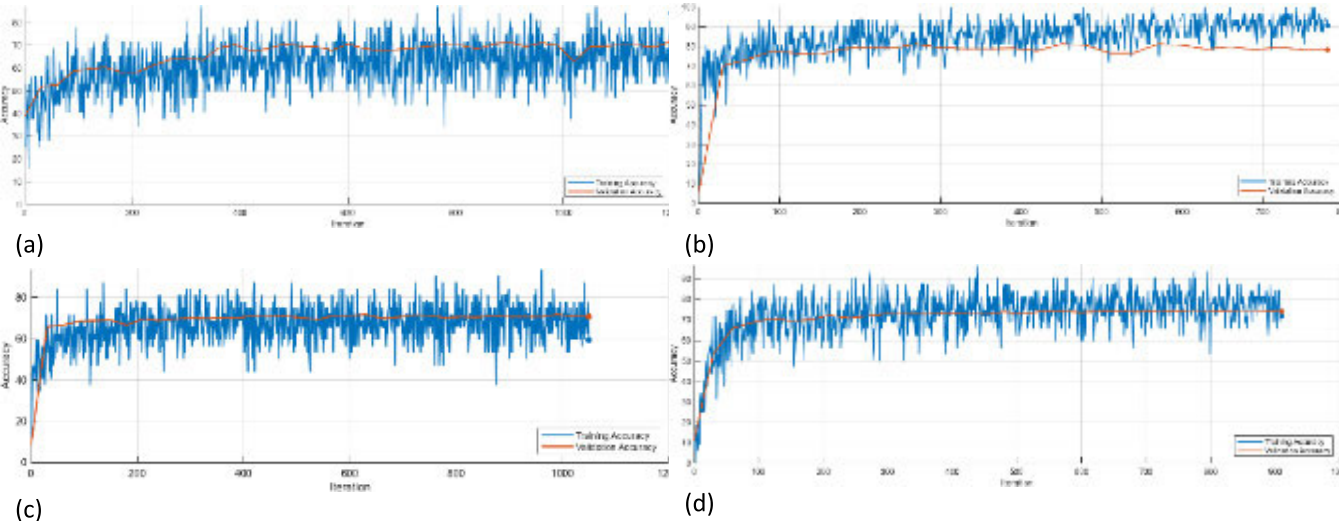


FIGURE 7. Accuracy curves for both models over APTOS dataset: (a) CNN with ADAM, (b) ResNet-50 with ADAM, (c) CNN with SGDM, and (d) ResNet-50 with SGDM.

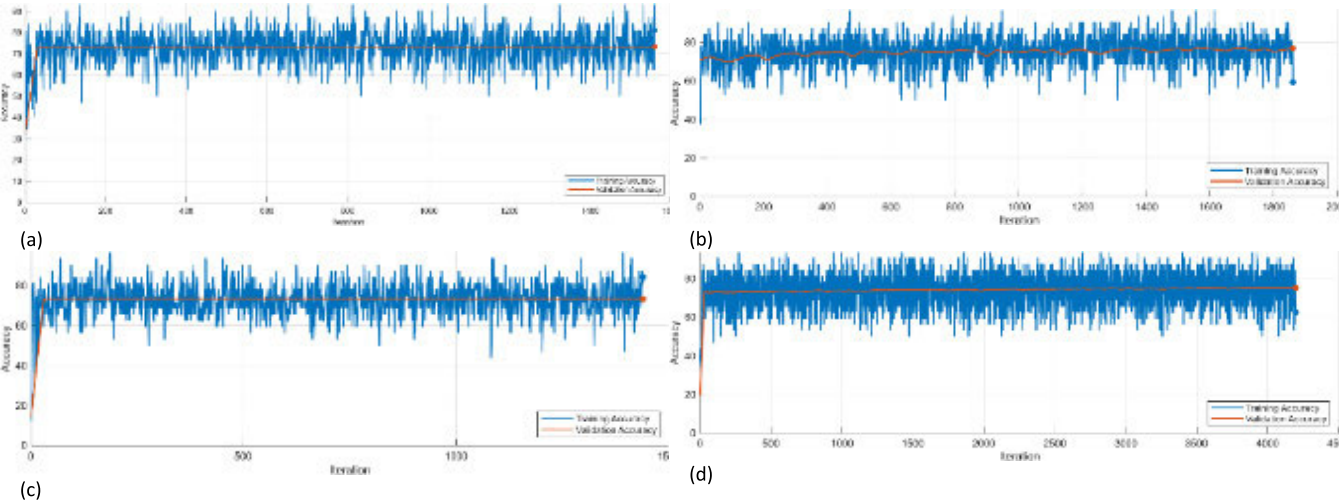


FIGURE 8. Accuracy curves for both models over EyePACS dataset: (a) CNN with ADAM, (b) ResNet-50 with ADAM, (c) CNN with SGDM, and (d) ResNet-50 with SGDM.

TABLE 3. The five-performance metrics of both models over APTOS dataset.

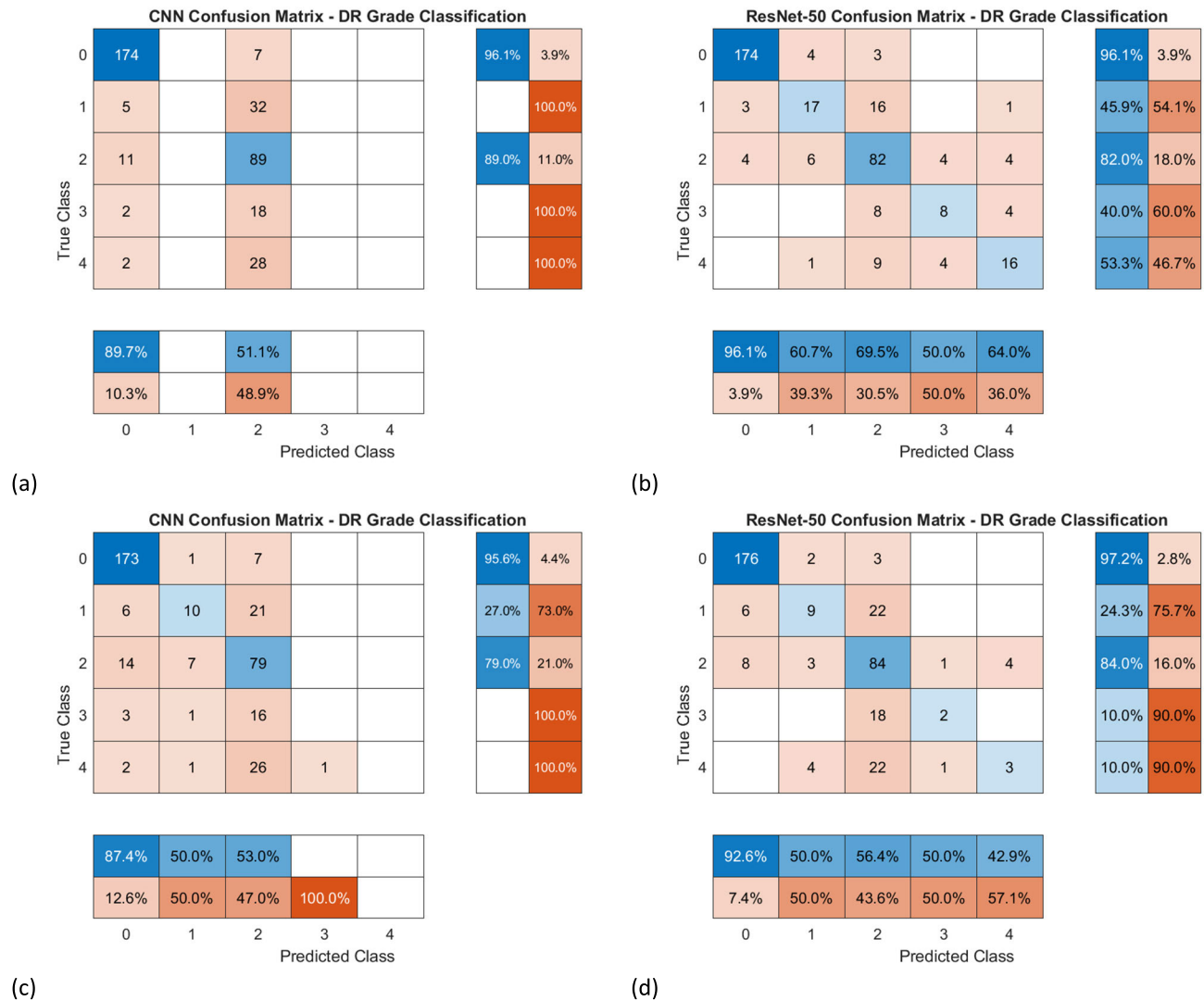
| Model     | Optimizer | Accuracy (%) | AUC    | F1-score | Precision | Recall |
|-----------|-----------|--------------|--------|----------|-----------|--------|
| CNN       | ADAM      | 75.43        | 0.9102 | 0.74     | 0.73      | 0.72   |
| CNN       | SGDM      | 72.19        | 0.8945 | 0.70     | 0.68      | 0.69   |
| ResNet-50 | ADAM      | 80.71        | 0.9539 | 0.79     | 0.78      | 0.78   |
| ResNet-50 | SGDM      | 78.62        | 0.9365 | 0.77     | 0.75      | 0.76   |

3. Hardware: All training was performed on GPU-enabled systems to speed up computation.

3) PERFORMANCE EVALUATION

Model performance was evaluated using five metrics:

1. Accuracy: Proportion of correct predictions.
2. AUC (Area Under Curve): Measures the model’s ability to distinguish between DR severity classes.
3. F1-Score: Harmonic mean of precision and recall, reflecting balance between false positives and false negatives.
4. Precision: Measures the proportion of correct positive predictions.



**FIGURE 9.** Confusion matrices for both models over APTOS dataset: (a) CNN with ADAM, (b) ResNet-50 with ADAM, (c) CNN with SGDM, and (d) ResNet-50 with SGDM.

**TABLE 4.** The five-performance metrics of both models over EyePACS dataset.

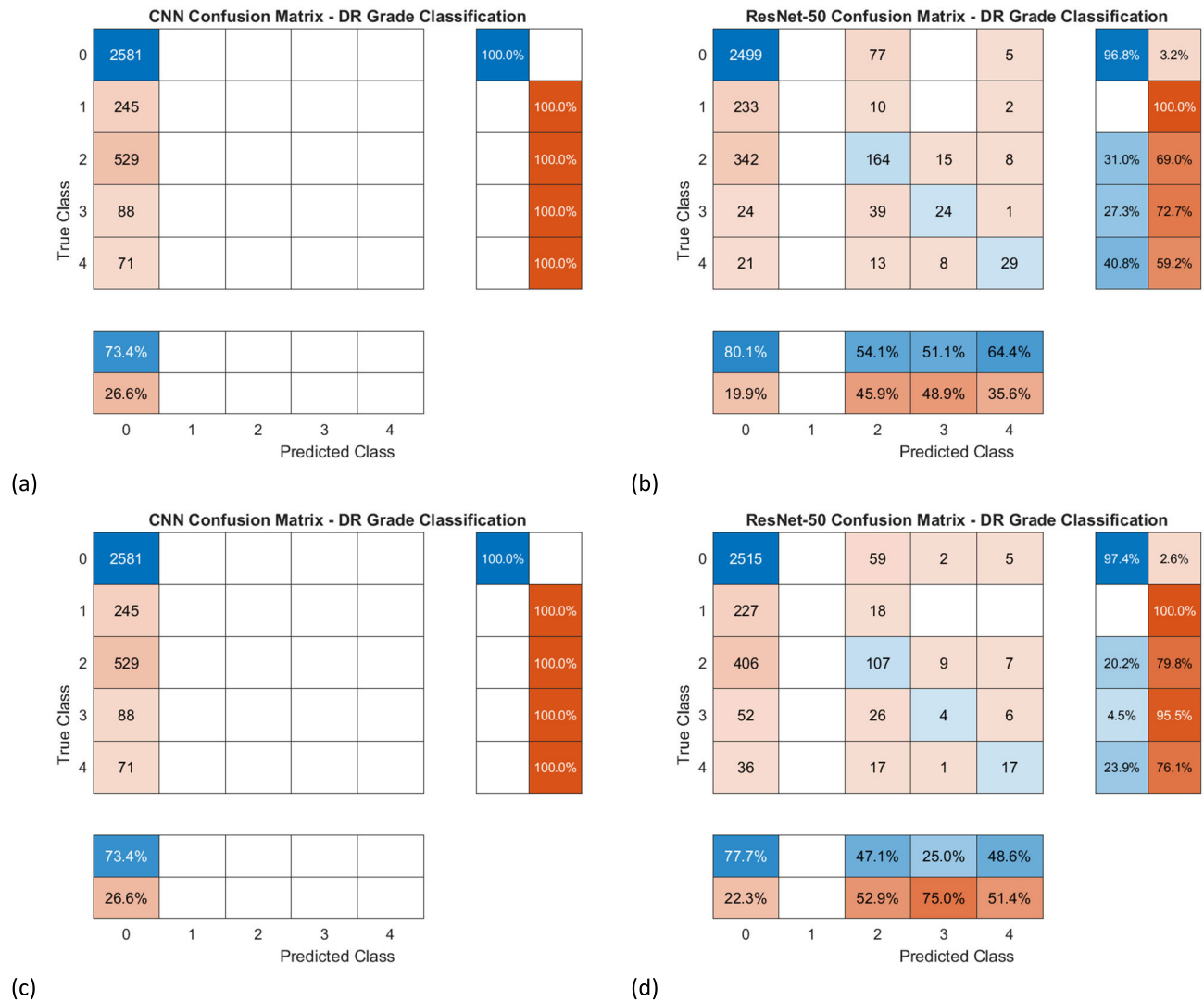
| Model     | Optimizer | Accuracy (%) | AUC    | F1-score | Precision | Recall |
|-----------|-----------|--------------|--------|----------|-----------|--------|
| CNN       | ADAM      | 68.45        | 0.8873 | 0.66     | 0.65      | 0.64   |
| CNN       | SGDM      | 65.72        | 0.8698 | 0.63     | 0.62      | 0.61   |
| ResNet-50 | ADAM      | 76.18        | 0.9251 | 0.74     | 0.72      | 0.73   |
| ResNet-50 | SGDM      | 74.06        | 0.9117 | 0.72     | 0.70      | 0.71   |

5. Recall (Sensitivity): Measures the proportion of actual positives correctly identified.

Each model-optimizer combination was evaluated independently on both datasets, and results were compared to determine the best-performing configuration.

### III. RESULTS

This section presents the performance results of both deep learning models, CNN and ResNet-50, trained using the ADAM and SGDM optimizers. The evaluation was conducted on the APTOS and EyePACS datasets using the metrics described earlier.



**FIGURE 10.** Confusion matrices for both models over EyePACS dataset: (a) CNN with ADAM, (b) ResNet-50 with ADAM, (c) CNN with SGDM, and (d) ResNet-50 with SGDM.

### A. TRAINING ACCURACY CURVES

Training and validation accuracy curves were used to monitor model learning behavior. Figures 7 and 8 show the accuracy curves for both models with ADAM and SGDM optimizers.

ResNet-50 with ADAM achieved faster convergence and higher stability compared to other configurations. CNN with SGDM required more epochs to reach similar accuracy levels but showed better generalization than CNN with ADAM. These results confirm that the optimizer choice directly affects model convergence speed and final accuracy.

### B. RESULTS ON APTOS DATASET

Table 3 summarizes the performance of both models on the APTOS dataset.

ResNet-50 with ADAM achieved the best results, reaching 80.71% accuracy and an AUC of 0.9539. The high F1-score

indicates stronger balance between precision and recall. CNN achieved lower performance due to its limited depth and lower feature extraction capacity.

Figure 9 shows confusion matrices for both models. Misclassifications mainly occurred between adjacent DR grades, such as moderate and severe, due to subtle differences in retinal features.

### C. RESULTS ON EYEPACS DATASET

Table 4 summarizes the performance of both models on the EyePACS dataset.

Performance on EyePACS was slightly lower than on APTOS because of higher image variability and noise. Still, ResNet-50 with ADAM maintained superior results across all metrics. This consistency across datasets confirms the generalization strength of transfer learning and adaptive optimization.

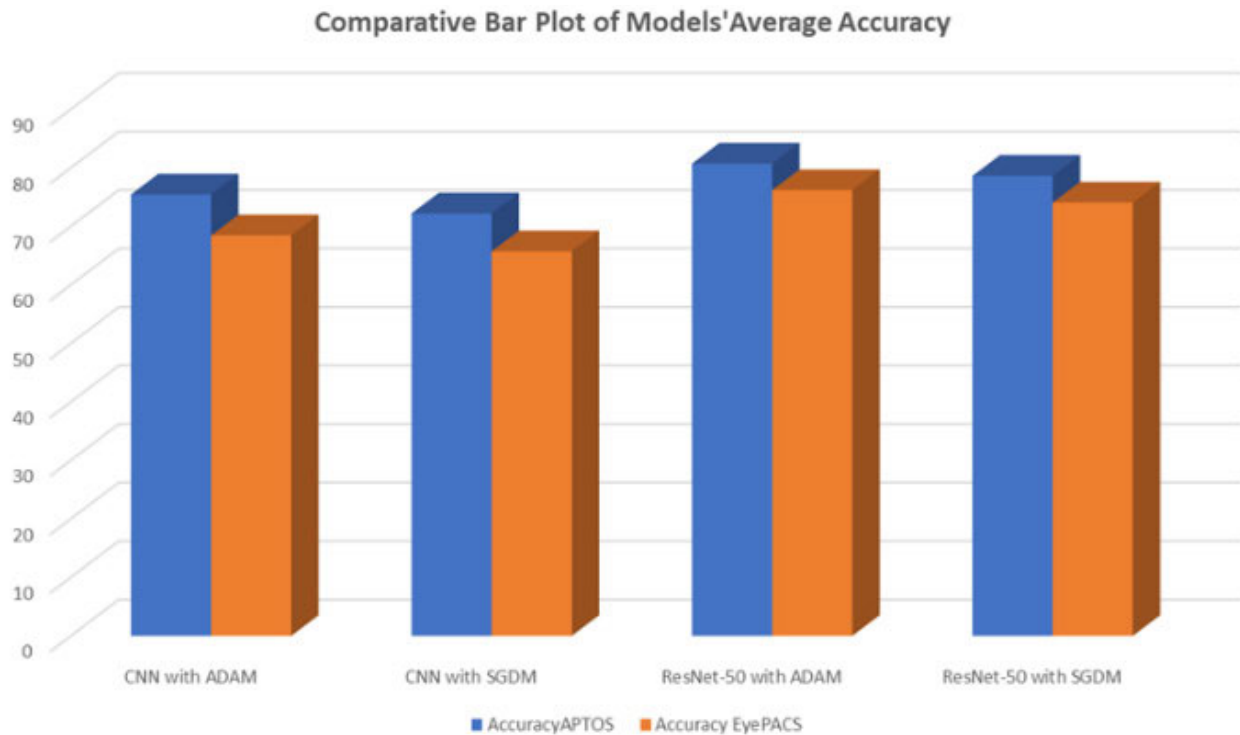


FIGURE 11. Models' average accuracy across both datasets.

Figure 10 shows confusion matrices for both models.

#### D. COMPARATIVE ANALYSIS

Figure 11 compares the models' average accuracy across both datasets. ResNet-50 with ADAM outperformed all other configurations. It consistently achieved high, showing effective feature learning and better optimizer adaptability.

The CNN model trained with SGDM showed stable learning but underperformed in feature discrimination, especially for intermediate DR grades. The ADAM optimizer improved CNN's performance slightly, but ResNet-50 benefited more due to its residual learning mechanism and adaptive parameter tuning.

#### IV. DISCUSSION

The results demonstrate that both architecture depth and optimizer choice significantly influence DR classification performance. The ResNet-50 model, with its skip connections, captured complex retinal features better than the CNN. ADAM provided faster convergence and more stable gradient updates than SGDM.

The performance difference between datasets reflects the importance of data diversity. The APTOS dataset, which contains higher-quality and better-centered images, produced stronger model accuracy. The EyePACS dataset, although larger, includes more variable illumination and focus, which affected feature extraction consistency.

Based on the confusion metrics, the model could misclassify adjacent DR grades such as mild and moderate DR cases, as well as moderate and severe, likely due to the lesions in these stages share similar visual patterns. Also, the model may produce errors between severe and proliferative DR for images with low contrast or limited visibility of neovascularization.

Compared to prior work, Gulshan et al. [9] achieved an AUC of 0.99 using a CNN on a large dataset, while Gargeya and Leng [10] reported an AUC of 0.94. In this study, ResNet-50 with ADAM achieved an AUC of 0.9539 on APTOS and 0.9251 on EyePACS, which are competitive results obtained with fewer images. This supports the effectiveness of transfer learning when combined with strong preprocessing and adaptive optimization.

Preprocessing played a key role in improving model accuracy. Applying CLAHE enhanced local contrast and helped the models detect lesions such as microaneurysms, hemorrhages, and exudates. Data augmentation and class balancing also reduced overfitting and improved the detection of higher DR grades.

The use of multiple evaluation metrics provided a comprehensive view of performance. Accuracy alone is not sufficient for medical diagnosis, especially in imbalanced datasets. The inclusion of F1-score, and AUC ensured that both precision and sensitivity were assessed fairly.

## A. IMPLICATIONS

The study confirms that transfer learning with deep residual networks is effective for DR detection and grading. Automated screening systems using such models can assist ophthalmologists in identifying high-risk patients early and reduce the workload in clinical settings. This approach supports large-scale screening programs, especially in regions with limited access to specialized eye care.

## B. LIMITATIONS AND FUTURE WORK

This research was limited to two datasets and two optimizers. Future work may include additional datasets, such as Messidor and DIARETDB1, to test model generalization further. Testing other advanced optimizers such as RMSProp and AdamW may also improve performance stability. Incorporating attention mechanisms or transformer-based architectures could help models focus on relevant retinal regions, improving interpretability and classification precision.

## V. CONCLUSION

ResNet-50 trained with the ADAM optimizer achieved the highest performance in classifying Diabetic Retinopathy severity from retinal fundus images. The model reached 80.71% accuracy and an AUC of 0.9539 on the APTOS dataset. The results demonstrate that combining transfer learning, adaptive optimization, and effective preprocessing produces accurate and reliable DR screening models. This approach supports the development of automated systems that can enhance early detection and help prevent vision loss among diabetic patients.

The study offers two contributions that extend previous comparative work. First, evaluating CNN and ResNet-50 across two large public datasets under identical preprocessing, class balancing, and training conditions. This would give readers a clear view of how each architecture behaves when exposed to real clinical variability in fundus images.

Second, the study framed the comparison around performance requirements linked to clinical screening by reporting accuracy, AUC, F1, recall, and precision because these metrics reflect the level of reliability needed during early DR detection. Also, the study highlighted the strength of ResNet-50 with ADAM in identifying both healthy and diseased cases, which supports its use in automated screening workflows.

The results pointed to a model-optimizer combination that offers consistent grading performance and stable behavior across two datasets. This helps developers select an architecture that supports clinical triage systems rather than focusing on algorithmic benchmarking alone.

## ACKNOWLEDGMENT

The authors extend their appreciation to Prince Sattam bin Abdulaziz University for funding this research work through the project number (PSAU/2025/03/33270).

## REFERENCES

- [1] J. W. Y. Yau et al., "Global prevalence and major risk factors of diabetic retinopathy," *Diabetes Care*, vol. 35, no. 3, pp. 556–564, Mar. 2012, doi: [10.2337/dc11-1909](#).
- [2] M. D. Abramoff, M. K. Garvin, and M. Sonka, "Retinal imaging and image analysis," *IEEE Rev. Biomed. Eng.*, vol. 3, pp. 169–208, 2010, doi: [10.1109/RBME.2010.2084567](#).
- [3] K. A. Alqurashi, K. S. Aljabri, and S. A. Bokhari, "Prevalence of diabetes mellitus in a Saudi community," *Ann. Saudi Med.*, vol. 31, no. 1, pp. 19–23, Jan. 2011, doi: [10.4103/0256-4947.75773](#).
- [4] K. Al-Rubeaan, A. M. Abu El-Asrar, A. M. Youssef, S. N. Subhani, N. A. Ahmad, A. H. Al-Sharqawi, A. Alguwaihes, M. S. Alotaibi, A. Al-Ghamdi, and H. M. Ibrahim, "Diabetic retinopathy and its risk factors in a society with a type 2 diabetes epidemic: A Saudi national diabetes registry-based study," *Acta Ophthalmologica*, vol. 93, no. 2, pp. e140–e147, Mar. 2015, doi: [10.1111/aos.12532](#).
- [5] K. M. Alabdulwahhab, "Diabetic retinopathy screening using non-mydratic fundus camera in primary health care settings—A multicenter study from Saudi Arabia," *Int. J. Gen. Med.*, vol. Volume 16, pp. 2255–2262, Jun. 2023, doi: [10.2147/ijgm.s410197](#).
- [6] A. B. Hansen, N. V. Hartvig, M. S. Jensen, K. Borch-Johnsen, H. Lund-Andersen, and M. Larsen, "Diabetic retinopathy screening using digital non-mydratic fundus photography and automated image analysis," *Acta Ophthalmologica Scandinavica*, vol. 82, no. 6, pp. 666–672, Dec. 2004, doi: [10.1111/j.1600-0420.2004.00350.x](#).
- [7] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, May 2015, doi: [10.1038/nature14539](#).
- [8] A. Esteva, B. Kuprel, R. A. Novoa, J. Ko, S. M. Swetter, H. M. Blau, and S. Thrun, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, Feb. 2017, doi: [10.1038/nature21056](#).
- [9] V. Gulshan, L. Peng, M. Coram, M. C. Stumpe, D. Wu, A. Narayanaswamy, S. Venugopalan, K. Widner, T. Madams, J. Cuadros, R. Kim, R. Raman, P. C. Nelson, J. L. Mega, and D. R. Webster, "Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs," *JAMA*, vol. 316, no. 22, pp. 2402–2410, Dec. 2016, doi: [10.1001/jama.2016.17216](#).
- [10] R. Gargeya and T. Leng, "Automated identification of diabetic retinopathy using deep learning," *Ophthalmology*, vol. 124, no. 7, pp. 962–969, Jul. 2017, doi: [10.1016/j.ophtha.2017.02.008](#).
- [11] D. S. W. Ting et al., "Development and validation of a deep learning system for diabetic retinopathy and related eye diseases using retinal images from multiethnic populations with diabetes," *JAMA*, vol. 318, no. 22, pp. 2211–2223, Dec. 2017, doi: [10.1001/jama.2017.18152](#).
- [12] H. Pratt, F. Coenen, D. M. Broadbent, S. P. Harding, and Y. Zheng, "Convolutional neural networks for diabetic retinopathy," *Proc. Comput. Sci.*, vol. 90, pp. 200–205, Jul. 2016, doi: [10.1016/j.procs.2016.07.014](#).
- [13] R. Rajalakshmi, S. Arulmalar, M. Usha, V. Prathiba, K. S. Kareemuddin, R. M. Anjana, and V. Mohan, "Validation of smartphone based retinal photography for diabetic retinopathy screening," *PLoS ONE*, vol. 10, no. 9, Sep. 2015, Art. no. e0138285, doi: [10.1371/journal.pone.0138285](#).
- [14] Y. Bilong, J.-C. Katte, G. Koki, G. Kagmeni, O. P. N. Obama, H. R. N. Fofe, C. Mvilongo, O. Nkengfack, A. M. Bimbai, E. Sobngwi, W. Mbacham, J. C. Mbanya, L. A. Bella, and A. Sharma, "Validation of smartphone-based retinal photography for diabetic retinopathy screening," *Ophthalmic Surgery, Lasers Imag. Retina*, vol. 50, no. S1, pp. S18–S22, May 2019, doi: [10.3928/23258160-20190108-05](#).
- [15] G. Ryu, K. Lee, D. Park, S. H. Park, and M. Sagong, "A deep learning model for identifying diabetic retinopathy using optical coherence tomography angiography," *Sci. Rep.*, vol. 11, no. 1, Nov. 2021, Art. no. 23024, doi: [10.1038/s41598-021-02479-6](#).
- [16] Ch. U. Kumari, A. Hemanth, V. Anand, D. S. Kumar, R. Naga Sanjeev, and T. S. Sri Harshitha, "Deep learning based detection of diabetic retinopathy using retinal fundus images," in *Proc. 3rd Int. Conf. Intell. Comput. Instrum. Control Technol. (ICICT)*, Kannur, India, Aug. 2022, pp. 1312–1316, doi: [10.1109/ICICT54557.2022.9917709](#).
- [17] S. V. Hemanth, S. Alagarsamy, and T. D. Rajkumar, "A novel deep learning model for diabetic retinopathy detection in retinal fundus images using pre-trained CNN and HWBLSTM," *J. Biomolecular Struct. Dyn.*, vol. 43, no. 13, pp. 6455–6473, Feb. 2024, doi: [10.1080/07391102.2024.2314269](#).

- [18] M. Touati, R. Touati, L. Nana, F. Benzarti, and S. Ben Yahia, "DRCCT: Enhancing diabetic retinopathy classification with a compact convolutional transformer," *Big Data Cognit. Comput.*, vol. 9, no. 1, p. 9, Jan. 2025, doi: [10.3390/bdcc9010009](https://doi.org/10.3390/bdcc9010009).
- [19] C. Lam, C. Yu, L. Huang, and D. Rubin, "Retinal lesion detection with deep learning using image patches," *Investigative Ophthalmology Vis. Sci.*, vol. 59, no. 1, pp. 590–596, Jan. 2018, doi: [10.1167/iov.17-22721](https://doi.org/10.1167/iov.17-22721).
- [20] Karthik, Maggie, and S. Dane. (2019). *APTOS 2019 Blindness Detection*. [Online]. Available: <https://kaggle.com/competitions/aptos2019-blindness-detection>
- [21] E. Dugas, Jared, Jorge, and W. Cukierski. (2015). *Diabetic Retinopathy Detection*. [Online]. Available: <https://kaggle.com/competitions/diabetic-retinopathy-detection>



**NADER A. RAHMAN MOHAMED** (Member, IEEE) was born in Egypt, in 1973. He received the B.S., M.S., and Ph.D. degrees in systems and biomedical engineering from the Faculty of Engineering, Cairo University, in June 1996, June 2002, and June 2009, respectively. He was a Biomedical Engineering Consultant in the field of ophthalmology, medical lasers, and endoscopy with the Center for Advanced Software & Biomedical Engineering Consultation (CASBEC), Faculty of Engineering, Cairo University. He joined the Department of Biomedical Engineering, Faculty of Engineering, Misr University for Science and Technology (MUST), in September 2009. He joined the Department of Biomedical Technology, College of Applied Medical Sciences in Al-Kharj, Prince Sattam Bin Abdulaziz University (PSAU), in January 2025. His current research interests include image and signal processing, artificial intelligence, neural network, biomedical equipment design and control, and biomechanics and biomaterials. He is a member of IEEE Engineering in Medicine and Biology Society (EMBS).



**MOHAMED A. EL-SAKHAWY** received the Ph.D. degree from Al-Azhar University, Cairo, Egypt, in 2014. He is currently a Lecturer with the Department of Medical Laboratory, College of Applied Medical Sciences, Prince Sattam Bin Abdulaziz University, Al-Kharj, Saudi Arabia, and a Researcher with the Department of Medicinal and Aromatic Plants, Desert Research Center, Cairo, Egypt. He has several research articles published in peer-reviewed international journals. He also serves as a reviewer for different high-reputation journals.



**ABDEL RAHMAN SHAABAN** received the Ph.D. degree in systems and biomedical engineering from Cairo University. He is currently the Technical Director of IntComp Tech Intelligence. His expertise encompasses artificial intelligence, digital transformation, IT governance, and cybersecurity. Alongside his industry leadership, he is also an Assistant Professor with El Shorouk Academy, where he contributes to research and teaching in biomedical engineering and artificial intelligence, fostering innovation at the intersection of academia and industry.

...